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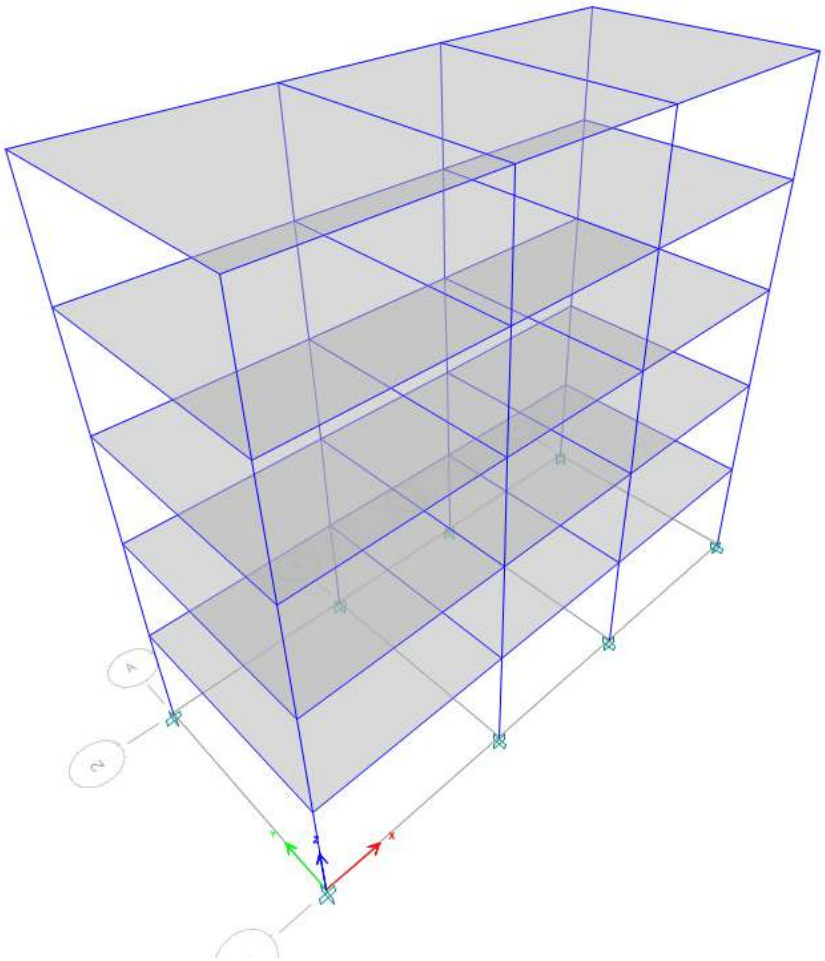

Project Report

Model File: G+4_BUILDING, Revision 0
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1 Structure Data

This chapter provides model geometry information, including items such as story levels, point coordinates, and element connectivity.

1.1 Story Data

Table 1.1 - Story Definitions

Tower	Name	Height m	Master Story	Similar To	Splice Story	Color
T1	Story5	3	Yes	None	No	Blue
T1	Story4	3	No	Story5	No	Green
T1	Story3	3	No	Story5	No	Cyan
T1	Story2	3	No	Story5	No	Red
T1	Story1	3	No	Story5	No	Magenta

1.2 Grid Data

Table 1.2 - Grid Definitions - General

Tower	Name	Type	Ux m	Uy m	Rz deg	Story Range	Bubble Size mm	Color
T1	G1	Cartesian	0	0	0	Default	1250	Gray6

Table 1.3 - Grid Definitions - Grid Lines

Name	Grid Line Type	ID	Ordinate m	Bubble Location	Visible
G1	X (Cartesian)	A	0	End	Yes
G1	X (Cartesian)	B	5.5	End	Yes
G1	X (Cartesian)	C	10	End	Yes
G1	X (Cartesian)	D	15.5	End	Yes
G1	Y (Cartesian)	1	0	Start	Yes
G1	Y (Cartesian)	2	6.5	Start	Yes

1.3 Point Coordinates

Table 1.4 - Point Bays

Label	Is Auto Point	X m	Y m	DZBelow m
1	No	0	0	0
2	No	0	6.5	0
3	No	5.5	0	0
4	No	5.5	6.5	0
5	No	10	0	0
6	No	10	6.5	0
7	No	15.5	0	0
8	No	15.5	6.5	0

1.4 Line Connectivity

Table 1.5 - Column Bays

Label	PointBayI	PointBayJ	IEndStory
C1	1	1	Below
C2	2	2	Below
C3	3	3	Below
C4	4	4	Below
C5	5	5	Below
C6	6	6	Below
C7	7	7	Below
C8	8	8	Below

Table 1.6 - Beam Bays

Label	PointBayI	PointBayJ
B1	1	2
B2	3	4
B3	5	6
B4	7	8
B5	1	3
B6	3	5
B7	5	7
B8	2	4
B9	4	6
B10	6	8

1.5 Area Connectivity

Table 1.7 - Floor Bays

Label	NumPoints	PointNumber	PointBay
F1	4	1	1
F1		2	3
F1		3	4
F1		4	2
F2	4	1	3
F2		2	5
F2		3	6
F2		4	4
F3	4	1	5
F3		2	7
F3		3	8
F3		4	6

1.6 Mass

Table 1.8 - Mass Source Definition

Name	Is Default	Include Lateral Mass?	Include Vertical Mass?	Lump Mass?	Source Self Mass?	Source Added Mass?	Source Load Patterns?	Move Mass Centroid?
MsSrc1	Yes	Yes	No	Yes	Yes	Yes	No	No

Table 1.9 - Mass Summary by Story

Story	UX kg	UY kg	UZ kg
Story5	45534.09	45534.09	0
Story4	49280.44	49280.44	0
Story3	49280.44	49280.44	0
Story2	49280.44	49280.44	0
Story1	49280.44	49280.44	0
Base	3746.35	3746.35	0

Table 1.10 - Mass Summary by Group

Group	Self Mass kg	Mass X kg	Mass Y kg	Mass Z kg
All	246402.18	246402.18	246402.18	0

2 Properties

This chapter provides property information for materials, frame sections, shell sections, and links.

2.1 Materials

Table 2.1 - Material Properties - General

Material	Type	SymType	Grade	Color	Notes
DefaultTendon$_{fpk1860}$	Tendon	Uniaxial	Grade fpk1860	Gray8Dark	
HYSD415	Rebar	Uniaxial	HYSD Grade 415	Yellow	
HYSD500	Rebar	Uniaxial	HYSD Grade 500	Green	
M20	Concrete	Isotropic	M20	Blue	

2.2 Frame Sections

Table 2.2 - Frame Section Property Definitions - Summary (Part 1 of 3)

Name	Material	Shape	Color	Area cm2	J cm4	I33 cm4	I22 cm4	I23 cm4	As2 cm2	As3 cm2
BEAM	M20	Concrete Rectangular	Magenta	805	84296.7	82177.1	35487.1	0	670.8	670.8
COLUMN	M20	Concrete Rectangular	Gray8Dark	1225	211416.1	125052.1	125052.1	0	1020.8	1020.8

Table 2.2 - Frame Section Property Definitions - Summary (Part 2 of 3)

Name	S33Pos cm3	S33Neg cm3	S22Pos cm3	S22Neg cm3	Z33 cm3	Z22 cm3	R33 mm	R22 mm	CG Offset 3 mm	CG Offset 2 mm	PNA Offset 3 mm	PNA Offset 2 mm
BEAM	4695.8	4695.8	3085.8	3085.8	7043.8	4628.8	101	66.4	0	0	0	0
COLUMN	7145.8	7145.8	7145.8	7145.8	10718.8	10718.8	101	101	0	0	0	0

Table 2.2 - Frame Section Property Definitions - Summary (Part 3 of 3)

Name	Area Modifier	As2 Modifier	As3 Modifier	J Modifier	I33 Modifier	I22 Modifier	Mass Modifier	Weight Modifier
BEAM	1	1	1	1	1	1	1	1
COLUMN	1	1	1	1	1	1	1	1

2.3 Shell Sections

Table 2.3 - Area Section Property Definitions - Summary

Name	Type	Element Type	Material	Total Thickness mm	Deck Material	Deck Depth mm
Slab150	Slab	Shell-Thin	M20	120		

2.4 Reinforcement Sizes

Table 2.4 - Reinforcing Bar Sizes

Name	Diameter mm	Area cm2
18	18	2.5

3 Loads

This chapter provides loading information as applied to the model.

3.1 Load Patterns

Table 3.1 - Load Pattern Definitions

Name	Is Auto Load	Type	Self Weight Multiplier
~LLRF	Yes	Other	0
Dead	No	Dead	1
Live	No	Live	0

3.2 Load Cases

Table 3.2 - Load Case Definitions - Summary

Name	Type
Dead	Linear Static
Live	Linear Static
Modal	Modal - Eigen

3.3 Load Combinations

Table 3.3 - Load Combination Definitions

Name	Type	Is Auto	Load Name	SF	Notes
DConS1	Linear Add	Yes	Dead	1.5	Dead [Strength]
DConS2	Linear Add	Yes	Dead	1.5	Dead + Live [Strength]
DConS2			Live	1.5	

4 Analysis Results

This chapter provides analysis results.

4.1 Structure Results

Table 4.1 - Base Reactions

Output Case	Case Type	FX kN	FY kN	FZ kN	MX kN-m	MY kN-m	MZ kN-m	X m	Y m	Z m
Dead	LinStatic	-48	-48	10301.38	33791.485	-80243.695	-264	0	0	0
Live	LinStatic	0	0	1511.25	4911.5625	-11712.1875	0	0	0	0
DConS1	Combination	-72	-72	15452.07	50687.2275	-120365.5425	-396	0	0	0
DConS2	Combination	-72	-72	17718.945	58054.5712	-137933.8237	-396	0	0	0

4.2 Story Results

Table 4.2 - Story Drifts

Story	Output Case	Case Type	Direction	Drift	Drift/	Label	X m	Y m	Z m
Story5	Dead	LinStatic	X	0.000171	1/5842	1	0	0	15
Story5	Dead	LinStatic	Y	0.000182	1/5485	7	15.5	0	15
Story5	Live	LinStatic	Y	4E-06	1/229594	5	10	0	15
Story5	DConS1	Combination	X	0.000257	1/3895	1	0	0	15
Story5	DConS1	Combination	Y	0.000273	1/3657	7	15.5	0	15
Story5	DConS2	Combination	X	0.000261	1/3830	1	0	0	15
Story5	DConS2	Combination	Y	0.000279	1/3589	7	15.5	0	15
Story4	Dead	LinStatic	X	0.000265	1/3769	7	15.5	0	12
Story4	Dead	LinStatic	Y	0.00032	1/3123	8	15.5	6.5	12
Story4	Live	LinStatic	Y	1E-06	1/691329	6	10	6.5	12
Story4	DConS1	Combination	X	0.000398	1/2513	7	15.5	0	12
Story4	DConS1	Combination	Y	0.00048	1/2082	8	15.5	6.5	12
Story4	DConS2	Combination	X	0.000399	1/2505	7	15.5	0	12
Story4	DConS2	Combination	Y	0.000482	1/2074	8	15.5	6.5	12
Story3	Dead	LinStatic	X	0.000368	1/2715	1	0	0	9
Story3	Dead	LinStatic	Y	0.000465	1/2151	7	15.5	0	9
Story3	Live	LinStatic	Y	8.302E-08	1/0	7	15.5	0	9
Story3	DConS1	Combination	X	0.000552	1/1810	1	0	0	9
Story3	DConS1	Combination	Y	0.000697	1/1434	7	15.5	0	9
Story3	DConS2	Combination	X	0.000553	1/1810	1	0	0	9
Story3	DConS2	Combination	Y	0.000698	1/1434	7	15.5	0	9
Story2	Dead	LinStatic	X	0.000446	1/2241	1	0	0	6
Story2	Dead	LinStatic	Y	0.000562	1/1778	7	15.5	0	6
Story2	Live	LinStatic	X	1E-06	1/0	8	15.5	6.5	6
Story2	Live	LinStatic	Y	1E-06	1/870930	5	10	0	6
Story2	DConS1	Combination	X	0.000669	1/1494	1	0	0	6
Story2	DConS1	Combination	Y	0.000843	1/1186	7	15.5	0	6
Story2	DConS2	Combination	X	0.00067	1/1492	1	0	0	6
Story2	DConS2	Combination	Y	0.000845	1/1184	7	15.5	0	6
Story1	Dead	LinStatic	X	0.000318	1/3143	7	15.5	0	3
Story1	Dead	LinStatic	Y	0.000383	1/2608	8	15.5	6.5	3

Table 4.2 - Story Drifts (continued)

Story	Output Case	Case Type	Direction	Drift	Drift/	Label	X m	Y m	Z m
Story1	Live	LinStatic	X	1E-06	1/0	2	0	6.5	3
Story1	Live	LinStatic	Y	1E-06	1/0	6	10	6.5	3
Story1	DConS1	Combination	X	0.000477	1/2095	7	15.5	0	3
Story1	DConS1	Combination	Y	0.000575	1/1739	8	15.5	6.5	3
Story1	DConS2	Combination	X	0.000478	1/2091	7	15.5	0	3
Story1	DConS2	Combination	Y	0.000576	1/1735	8	15.5	6.5	3

Table 4.3 - Story Max Over Avg Drifts

Story	Output Case	Case Type	Direction	Max Drift mm	Avg Drift mm	Ratio
Story5	Dead	LinStatic	X	0.514	0.435	1.181
Story5	Dead	LinStatic	Y	0.547	0.409	1.336
Story5	Live	LinStatic	Y	0.013	0	2692000000000
Story5	DConS1	Combination	X	0.77	0.652	1.181
Story5	DConS1	Combination	Y	0.82	0.614	1.336
Story5	DConS2	Combination	X	0.783	0.652	1.201
Story5	DConS2	Combination	Y	0.836	0.614	1.362
Story4	Dead	LinStatic	X	0.796	0.722	1.102
Story4	Dead	LinStatic	Y	0.961	0.795	1.208
Story4	Live	LinStatic	Y	0.004	0	1660000000000
Story4	DConS1	Combination	X	1.194	1.083	1.102
Story4	DConS1	Combination	Y	1.441	1.193	1.208
Story4	DConS2	Combination	X	1.198	1.083	1.105
Story4	DConS2	Combination	Y	1.447	1.193	1.213
Story3	Dead	LinStatic	X	1.105	1.016	1.087
Story3	Dead	LinStatic	Y	1.395	1.183	1.179
Story3	Live	LinStatic	Y	0.000249	0	65307318489
Story3	DConS1	Combination	X	1.657	1.524	1.087
Story3	DConS1	Combination	Y	2.092	1.775	1.179
Story3	DConS2	Combination	X	1.658	1.524	1.087
Story3	DConS2	Combination	Y	2.093	1.775	1.179
Story2	Dead	LinStatic	X	1.338	1.222	1.095
Story2	Dead	LinStatic	Y	1.687	1.423	1.186
Story2	Live	LinStatic	X	0.002	0	3534000000000
Story2	Live	LinStatic	Y	0.003	0	912800000000
Story2	DConS1	Combination	X	2.008	1.833	1.095
Story2	DConS1	Combination	Y	2.53	2.134	1.186
Story2	DConS2	Combination	X	2.011	1.833	1.097
Story2	DConS2	Combination	Y	2.534	2.134	1.188
Story1	Dead	LinStatic	X	0.954	0.872	1.094
Story1	Dead	LinStatic	Y	1.15	0.969	1.187
Story1	Live	LinStatic	X	0.002	0	4584000000000
Story1	Live	LinStatic	Y	0.003	0	3318000000000
Story1	DConS1	Combination	X	1.432	1.309	1.094

Table 4.3 - Story Max Over Avg Drifts (continued)

Story	Output Case	Case Type	Direction	Max Drift mm	Avg Drift mm	Ratio
Story1	DConS1	Combination	Y	1.726	1.453	1.187
Story1	DConS2	Combination	X	1.435	1.309	1.096
Story1	DConS2	Combination	Y	1.729	1.453	1.19

Table 4.4 - Story Forces

Story	Output Case	Case Type	Location	P kN	VX kN	VY kN	T kN-m	MX kN-m	MY kN-m
Story5	Dead	LinStatic	Top	1988.3977	-8	-8	-44	6454.2925	-15418.0822
Story5	Dead	LinStatic	Bottom	2061.876	-8	-8	-44	6717.097	-16011.539
Story5	Live	LinStatic	Top	302.25	0	0	0	982.3125	-2342.4375
Story5	Live	LinStatic	Bottom	302.25	0	0	0	982.3125	-2342.4375
Story5	DConS1	Combination	Top	2982.5966	-12	-12	-66	9681.4388	-23127.1233
Story5	DConS1	Combination	Bottom	3092.814	-12	-12	-66	10075.6455	-24017.3085
Story5	DConS2	Combination	Top	3435.9716	-12	-12	-66	11154.9076	-26640.7796
Story5	DConS2	Combination	Bottom	3546.189	-12	-12	-66	11549.1142	-27530.9647
Story4	Dead	LinStatic	Top	4050.2737	-16	-16	-88	13171.3895	-31429.6212
Story4	Dead	LinStatic	Bottom	4123.752	-16	-16	-88	13458.194	-32047.078
Story4	Live	LinStatic	Top	604.5	0	0	0	1964.625	-4684.875
Story4	Live	LinStatic	Bottom	604.5	0	0	0	1964.625	-4684.875
Story4	DConS1	Combination	Top	6075.4106	-24	-24	-132	19757.0843	-47144.4318
Story4	DConS1	Combination	Bottom	6185.628	-24	-24	-132	20187.291	-48070.617
Story4	DConS2	Combination	Top	6982.1606	-24	-24	-132	22704.0218	-54171.7443
Story4	DConS2	Combination	Bottom	7092.378	-24	-24	-132	23134.2285	-55097.9295
Story3	Dead	LinStatic	Top	6112.1497	-24	-24	-132	19912.4865	-47465.1602
Story3	Dead	LinStatic	Bottom	6185.628	-24	-24	-132	20223.291	-48106.617
Story3	Live	LinStatic	Top	906.75	0	0	0	2946.9375	-7027.3125
Story3	Live	LinStatic	Bottom	906.75	0	0	0	2946.9375	-7027.3125
Story3	DConS1	Combination	Top	9168.2246	-36	-36	-198	29868.7298	-71197.7403
Story3	DConS1	Combination	Bottom	9278.442	-36	-36	-198	30334.9365	-72159.9255
Story3	DConS2	Combination	Top	10528.3496	-36	-36	-198	34289.1361	-81738.709
Story3	DConS2	Combination	Bottom	10638.567	-36	-36	-198	34755.3427	-82700.8942
Story2	Dead	LinStatic	Top	8174.0257	-32	-32	-176	26677.5835	-63524.6992
Story2	Dead	LinStatic	Bottom	8247.504	-32	-32	-176	27012.388	-64190.156
Story2	Live	LinStatic	Top	1209	0	0	0	3929.25	-9369.75
Story2	Live	LinStatic	Bottom	1209	0	0	0	3929.25	-9369.75
Story2	DConS1	Combination	Top	12261.0386	-48	-48	-264	40016.3753	-95287.0488
Story2	DConS1	Combination	Bottom	12371.256	-48	-48	-264	40518.582	-96285.234
Story2	DConS2	Combination	Top	14074.5386	-48	-48	-264	45910.2503	-109341.6738
Story2	DConS2	Combination	Bottom	14184.756	-48	-48	-264	46412.457	-110339.859
Story1	Dead	LinStatic	Top	10235.9017	-40	-40	-220	33466.6805	-79608.2382
Story1	Dead	LinStatic	Bottom	10309.38	-40	-40	-220	33825.485	-80297.695
Story1	Live	LinStatic	Top	1511.25	0	0	0	4911.5625	-11712.1875
Story1	Live	LinStatic	Bottom	1511.25	0	0	0	4911.5625	-11712.1875
Story1	DConS1	Combination	Top	15353.8526	-60	-60	-330	50200.0208	-119412.3573

Table 4.4 - Story Forces (continued)

Story	Output Case	Case Type	Location	P kN	VX kN	VY kN	T kN-m	MX kN-m	MY kN-m
Story1	DConS1	Combination	Bottom	15464.07	-60	-60	-330	50738.2275	-120446.5425
Story1	DConS2	Combination	Top	17620.7276	-60	-60	-330	57567.3645	-136980.6385
Story1	DConS2	Combination	Bottom	17730.945	-60	-60	-330	58105.5712	-138014.8237

4.3 Modal Results

Table 4.5 - Modal Periods And Frequencies

Case	Mode	Period sec	Frequency cyc/sec	CircFreq rad/sec	Eigenvalue rad2/sec2
Modal	1	1	1	6.2834	39.4812
Modal	2	0.864	1.157	7.2696	52.8473
Modal	3	0.792	1.263	7.934	62.9489
Modal	4	0.306	3.27	20.5458	422.1289
Modal	5	0.272	3.677	23.1018	533.6912
Modal	6	0.245	4.075	25.6019	655.4579
Modal	7	0.161	6.205	38.9874	1520.021
Modal	8	0.15	6.682	41.9839	1762.647
Modal	9	0.132	7.595	47.7192	2277.1264
Modal	10	0.103	9.665	60.7295	3688.0692
Modal	11	0.1	10.011	62.9017	3956.6246
Modal	12	0.085	11.7	73.5121	5404.0243

Table 4.6 - Modal Participating Mass Ratios (Part 1 of 2)

Case	Mode	Period sec	UX	UY	UZ	SumUX	SumUY	SumUZ	RX	RY	RZ	SumRX
Modal	1	1	0	0.8124	0	0	0.8124	0	0.2089	0	0	0.2089
Modal	2	0.864	0.8258	0	0	0.8258	0.8124	0	0	0.194	0	0.2089
Modal	3	0.792	0	0	0	0.8258	0.8124	0	0	0	0.8173	0.2089
Modal	4	0.306	0	0.1113	0	0.8258	0.9237	0	0.5828	0	0	0.7917
Modal	5	0.272	0.1056	0	0	0.9314	0.9237	0	0	0.6154	0	0.7917
Modal	6	0.245	0	0	0	0.9314	0.9237	0	0	0	0.1082	0.7917
Modal	7	0.161	0	0.0463	0	0.9314	0.97	0	0.1014	0	0	0.8931
Modal	8	0.15	0.0423	0	0	0.9737	0.97	0	0	0.0916	0	0.8931
Modal	9	0.132	0	0	0	0.9737	0.97	0	0	0	0.0452	0.8931
Modal	10	0.103	0	0.0227	0	0.9737	0.9927	0	0.0849	0	0	0.978
Modal	11	0.1	0.0201	0	0	0.9938	0.9927	0	0	0.0804	0	0.978
Modal	12	0.085	0	0	0	0.9938	0.9927	0	0	0	0.0222	0.978

Table 4.6 - Modal Participating Mass Ratios (Part 2 of 2)

Case	Mode	SumRY	SumRZ
Modal	1	0	0
Modal	2	0.194	0
Modal	3	0.194	0.8173
Modal	4	0.194	0.8173
Modal	5	0.8094	0.8173
Modal	6	0.8094	0.9255

Table 4.6 - Modal Participating Mass Ratios (Part 2 of 2, continued)

Case	Mode	SumRY	SumRZ
Modal	7	0.8094	0.9255
Modal	8	0.9011	0.9255
Modal	9	0.9011	0.9707
Modal	10	0.9011	0.9707
Modal	11	0.9814	0.9707
Modal	12	0.9814	0.9928

Table 4.7 - Modal Load Participation Ratios

Case	ItemType	Item	Static %	Dynamic %
Modal	Acceleration	UX	99.99	99.38
Modal	Acceleration	UY	99.99	99.27
Modal	Acceleration	UZ	0	0

Table 4.8 - Modal Direction Factors

Case	Mode	Period sec	UX	UY	UZ	RZ
Modal	1	1	0	1	0	0
Modal	2	0.864	1	0	0	0
Modal	3	0.792	0	0	0	1
Modal	4	0.306	0	1	0	0
Modal	5	0.272	1	0	0	0
Modal	6	0.245	0	0	0	1
Modal	7	0.161	0	1	0	0
Modal	8	0.15	1	0	0	0
Modal	9	0.132	0	0	0	1
Modal	10	0.103	0	1	0	0
Modal	11	0.1	1	0	0	0
Modal	12	0.085	0	0	0	1